

Program No: 6

File Name:

Ex. No:

Date: _____

Simulate and mitigate common web vulnerabilities (XSS, SQL Injection) in a sample application

AIM:

To demonstrate common web vulnerabilities (XSS and SQL injection) safely and show simple mitigations in a browser-based demo.

ALGORITHM:

Step 1 : Create an HTML file with a form containing Name, Email, and Password inputs and a submit button.

Step 2 : Add CSS to style the form for readability and better appearance.

Step 3: Write JavaScript to validate inputs on submit: check required fields, validate email format, and check password length.

Step 4: Display error messages next to invalid fields if any validation fails.

Step 5: Show a success message when all inputs are valid.

Step 6: Test the form with valid and invalid data to ensure validation works correctly.

PROGRAM:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Practice 14: Web Vulnerabilities</title>
```

```
<style>
```

```
body { font-family: Arial, sans-serif; padding: 20px; background: #f4f4f4; }
```

```
h1 { color: #007acc; }
```

```
input, button { padding: 8px; margin: 5px 0; }
```

```
.output { margin-top: 10px; color: green; font-weight: bold; }
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Simulate and Mitigate Web Vulnerabilities</h1>
```

```
<!-- XSS Example -->
```

```
<h2>XSS Simulation</h2>
```

```
<input type="text" id="xssInput" placeholder="Enter your name">
```

```
<button onclick="showMessage()">Submit</button>
```

```
<div id="xssOutput" class="output"></div>
```

```
<!-- SQL Injection Example (Simulation) -->
```

```
<h2>SQL Injection Simulation</h2>
```

```
<input type="text" id="username" placeholder="Username">
```

```
<input type="text" id="password" placeholder="Password">
```

```
<button onclick="login()">Login</button>
```

```
<div id="sqlOutput" class="output"></div>
```

```
<script>
```

```
// --- XSS Vulnerable Version ---
```

```
function showMessage() {
```

```
    const input = document.getElementById('xssInput').value;
```

```
    // Vulnerable version (don't use in production)
```

```
    // document.getElementById('xssOutput').innerHTML = "Hello " + input;
```

```
    // Safe version (mitigation)
```

```
    const safeText = document.createTextNode(input);
```

```
    const output = document.getElementById('xssOutput');
```

```
    output.innerHTML = "";
```

```
    output.appendChild(document.createTextNode("Hello "));
```

```
    output.appendChild(safeText);
```

```
}
```

```
// --- SQL Injection Simulation ---
```

```
const users = [
```

```
    { username: 'admin', password: '1234' },
```

```
    { username: 'user', password: 'abcd' }
```

```
];
```

```
function login() {

    const username = document.getElementById('username').value;

    const password = document.getElementById('password').value;


    // Vulnerable simulation (for demonstration only)

    // const query = `SELECT * FROM users WHERE username='${username}' AND
password='${password}'`;

    // Safe mitigation using simple check (like prepared statements)

    const user = users.find(u => u.username === username && u.password === password);

    const output = document.getElementById('sqlOutput');

    if (user) {

        output.textContent = 'Login Successful!';

    } else {

        output.textContent = 'Invalid username or password.';

    }

}

</script>

</body>

</html>
```

OUTPUT:

← → ↻ ⓘ File C:/Users/admin/Documents/index.html

Simulate and Mitigate Web Vulnerabilities

XSS Simulation

Hello alice

SQL Injection Simulation

Login Successful!

RESULT:

Thus the program was successfully executed